

Style sample: the CS 312 example problem and quiz questions

Assignment handout / Prediction quiz / Quiz debrief

dl-alchemy (after Stanford CS 312: Deep Learning Alchemy)

2026-09-19

content verbatim from `deep-learning-alchemy.github.io`; layout only

1 What a handout looks like

An assignment scopes out a phenomenon and lists directions to explore. Each direction is an orange Problem box with a suggested GPU budget. The baseline code the diffs refer to is printed once, in a grey panel.

Problem (ablations): How much does each component improve loss? (budget set at calibration)

Start from the baseline depth 8 transformer that we pre-train on 614M tokens. This transformer (implementation provided below) features multiple components such as prenorm, attention, MLP, residuals, and positional embeddings. In this problem, we want to understand how much each component improves loss. Explore the following directions.

- (a) Perform leave-one-out ablations on the role of each component in our current model.
- (b) Understand the role of the residual connection. Can you come up with different schemes that improve the loss benefit of the residual connection?

Deliverable: Plots or tables in your report, with one sentence per result saying what you now believe.

Resource requirements: Stated per problem once the baseline run time on the GB10 is measured.

```
class Block(nn.Module):
    def __init__(self, width, ...):
        super().__init__()
        self.attn_norm = RMSNorm(width)
        self.attn = Attention(width, ...)
        self.mlp_norm = RMSNorm(width)
        self.mlp = MLP(width, ...)
    def update(self, x):
        attn = self.attn(self.attn_norm(x))
        mlp = self.mlp(self.mlp_norm(x + attn))
        return attn + mlp
    def forward(self, x):
        return x + self.update(x)
```

Low-Resource Tip: If the GPU window is short

Blue boxes carry tips: a reduced version of a sweep that still answers the question, or a known sharp edge of this machine.

2 What a quiz looks like

Each question is one held-out experiment that was actually run, given as a description and a code diff, with the difficulty in the header. The layout of [Section 1](#) carries over.

Question 1 (easy): Rank the ablations (2 points)

We train each of the following depth-8 transformer variants using the same training recipe and hyperparameters. Rank the variants from lowest to highest final validation loss. Use $<$, $>$, or $=$ between the letters; use $=$ when the losses are within 0.01.

A Baseline

B No prenorm

C No residual

D No MLP

E No attention

F No positional embeddings

No prenorm diff.

```
@@ -10,4 +10,4 @@ class Block.update
def update(self, x):
-   attn = self.attn(self.attn_norm(x))
+   attn = self.attn(x)
-   mlp = self.mlp(self.mlp_norm(x + attn))
+   mlp = self.mlp(x + attn)
    return attn + mlp
```

No residual diff.

```
@@ -16,2 +16,2 @@ class Block.forward
def forward(self, x):
-   return x + self.update(x)
+   return self.update(x)
```

Answer: _____

Question 2 (hard): Mixed residual (3 points)

We train the default 8 layer transformer as well as a variant where instead of having the residual connection come from the previous layer, we take 0.5 contribution from the previous layer and 0.5 contribution from two layers before (except for the first layer). For each method, we take the best loss over six learning rates $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$. What is your prediction of the loss difference between each method, $L_{\text{mixed residual}} - L_{\text{default}}$?

```

@@ -34,3 +34,9 @@ Transformer.forward
x_two_back = None
for i, block in enumerate(self.blocks):
-   x = block(x)
+   update = block.update(x)
+   residual = (
+       x if i == 0
+       else 0.5 * x + 0.5 * x_two_back
+   )
+   x_two_back, x = x, residual + update

```

$L_{\text{mixed residual}} - L_{\text{default}} =$ _____

3 What a debrief looks like

After the quiz, each question gets the measured outcome in a black box and the intuition that would have predicted it in a blue box.

Result: Question 2

{Measured losses of both methods at each learning rate, the best-of-grid difference, the seed-noise band, your answer, and the score.}

Mental model: {name of the intuition}

{Two or three sentences: the picture of the network that makes this outcome the expected one, and the nearby experiment where the same picture would predict the opposite.}

Variant	LR	Val loss	Δ vs baseline
$\langle name \rangle$	$\langle lr \rangle$	$\langle loss \rangle$	$\langle \Delta \rangle$

Table 1: Tables and figures are numbered and captioned below, as in the day-N handouts.